

6a	change in magnetic flux density $\Delta B = 5 - 50 = -45 \text{ mT}$ (from graph, accept positive value) average induced e.m.f. $= \Delta\Phi / \Delta t = \Delta NBA / \Delta t = NA \Delta B / \Delta t$ $= 180 \times \pi (5.3 \times 10^{-3} / 2)^2 \times (-45 \times 10^{-3}) / 0.30$ $= -5.957 \times 10^{-4} = -6.0 \times 10^{-4} \text{ V}$	B1 C1 A1
6b	As the coil moves away from the magnet, there is <u>a rate of change of magnetic flux linkage through the coil</u> due to the decreasing magnetic flux density <u>resulting in an induced e.m.f.</u> in the coil, according to Faraday's law. With the resistor connected in a closed loop with the coil, <u>a current is induced</u> in the <u>closed loop / coil</u> that opposes the motion of the coil, according to Lenz's law. As <u>work is done (by external force) / energy is required to move the coil away</u> at constant speed, this gives rise to the heating effect of induced current (in resistor).	B1 B1 B1

7ai	As p.d. becomes negative, there are photoelectrons that reach collector C, contributing	B1
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